

Dylan Xinming Hou

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Research Interests

My current research focuses on **human-AI interaction**, **human-centered evaluation & alignment**, and **AI trustworthiness and safety**, with a broader goal of advancing **AI for social good**.

I am especially interested in how to evaluate and align AI systems around human needs, values, and social contexts; how to build trustworthy AI systems that remain safe and useful in real-world interaction; and how to build **digital agents**, **multimodal systems**, and **mind-inspired (world) models** that support more natural, adaptive, and socially responsible human-AI collaboration.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

Abu Dhabi, UAE

M.S. STUDENT IN **MACHINE LEARNING**, ADVISED BY **GUS XIA** AT **MUSIC X LAB**

Aug. 2025 - 2027

- Current graduate study focused on machine learning, human-centered AI, human-AI interaction, and mind-inspired (world) models.

Gaoling School of Artificial Intelligence, Renmin University of China (RUC)

Beijing, China

BACHELOR OF ENGINEERING (B.ENG.) IN **ARTIFICIAL INTELLIGENCE** WITH HONORS

Sept. 2021 - Jun. 2025

- GPA: 3.68/4; Average score: 87.2.
- Relevant Courses: Machine Learning, Deep Learning, Natural Language Processing, Algorithm Design and Analysis, Data Structures and Algorithms, Python Programming, C & C++ Programming, Probability and Statistics, Advanced Algebra, Mathematical Analysis.

Publications

Detecting AI-Generated Video: A Vision-Language Dual-View Survey

Dylan Xinming Hou, Juntian Zhang, Xu Gu, Yichen Wu, Nils Lukas, Gus Xia, Xiuying Chen, Yuhan Liu.

ACL 2026 Findings [Project] [Paper] [Resources]

MuseMT: Clinically Grounded Affective Interaction Through Generative Music

Dylan Xinming Hou, Xin Shu, Zhuojian Cai, Yuting Wu, Yueqing Hu, Enkelejda Kasneci, Haiyang Geng, Gus Xia.

Under review, 2026

RuleEdit: Benchmarking Rule-Level Knowledge Editing in Large Language Models

Jizhi Zhang, **Dylan Xinming Hou**, Zhaoyi Li, Muhammad Asif Ali, Gus Xia, Lijie Hu.

Under review, 2025

Does Edited Knowledge Systematically Generalize in Multi-model Large Language Models?

Zhuoqi He, Zhaoyi Li, **Dylan Xinming Hou**, Daniil Ognev, Gus Xia, Lijie Hu.

Under review, 2025

CoAct: A Global-Local Hierarchy for Autonomous Agent Collaboration

Xinming Hou, Mingming Yang, Wenxiang Jiao, Xing Wang, Zhaopeng Tu, Wayne Xin Zhao.

Preprint, 2024 [arXiv] [Code]

Research Experience

Mohamed bin Zayed University of Artificial Intelligence

Abu Dhabi, UAE

RESEARCH ASSISTANT AT **MUSIC X LAB**, ADVISED BY **PROF. GUS XIA**

Aug. 2025 - Present

- **Research Topics:** human-AI interaction, human-centered evaluation & alignment, AI trustworthiness & safety, multimodal AI, LLM knowledge editing, generative music.
- Study clinically grounded affective interaction and generative music systems, connecting machine learning methods with human-facing therapeutic and interaction scenarios.
- Explore how to build digital agents and mind-inspired models that support more natural, adaptive, and socially responsible human-AI collaboration.

Renmin University of China

Beijing, China

RESEARCH ASSISTANT AT [RUC AI BOX](#), ADVISED BY [PROF. WAYNE XIN ZHAO](#)

Jul. 2022 - Jun. 2025

- **Research Topics:** machine learning, natural language processing, factuality control, prompt pipelines, interpretability.
- Worked on knowledge discernment and memory management methods to improve factuality control in language models.
- Built self-improvement and prompt-pipeline frameworks for complex cognitive and reasoning tasks, incorporating in-context learning methods.
- Explored Transformer mechanisms and multilingual reasoning behavior from a neuron-level and cognition-inspired perspective.

Tencent AI Lab

Shenzhen, China

RESEARCH INTERN AT [AI RESEARCH GROUP](#), ADVISED BY [WENXIANG JIAO](#) AND [XING WANG](#)

Jul. 2023 - Jun. 2024

- **Research Topics:** LLM-based agents, multi-agent collaboration, human-inspired organization.
- Explored the potential of transferring human societal work patterns to AI systems.
- Investigated efficient and flexible collaboration methods for autonomous agents to improve complex real-world task handling.

Project Portfolio (Selected)

Classical Chinese Poetry Creation Model

Beijing, China

FOUNDER & DEVELOPER, ADVISED BY [PROF. RUI YAN](#) AND [PROF. DANJUN CAI](#) / [\[PRE. SLIDES\]](#)

Feb. 2023 - Apr. 2024

- Proposed a Chinese Tang Poetry creation pre-trained language model, which endows the model with vitality and a complete life cycle like Tang dynasty poets, and enables emotional interaction with users.
- Won the Best Creativity Award in The 14th Student Entrepreneurship Star Competition, JD Cup.

Survey of Large Language Models

Beijing, China

PROOFREADER, ADVISED BY [PROF. WAYNE XIN ZHAO](#) / [\[PAPER\]](#) / [\[GITHUB\]](#) / [\[BOOK\]](#)

Mar. 2023 - May. 2023

- Conducted overall and detailed review to translate the paper into a high-quality Chinese version.
- Corrected the inconsistencies and ambiguities in the text and provided feasible suggestions for improvement.

Implementation of a Basic Search Engine

Beijing, China

FOUNDER & DEVELOPER, ADVISED BY [PROF. JIAXIN MAO](#) / [\[CODE\]](#)

Aug. 2022

- Coursework for Artificial Intelligence Integrated Design, voted as the best system (1/14)
- Built A website that supports school community network search using HTML5, CSS, and JS languages.
- Applied the advanced methods of web crawler and classical algorithms of information retrieval (such as TF-IDF, PAGE-RANK, etc.)

Skills

Research Human-AI interaction, human-centered AI, evaluation & alignment, trustworthiness & safety, AI for social good.

AI Topics LLMs, digital agents, multimodal learning, interpretability, affective computing, mind-inspired (world) models.

Programming Python, C/C++, $\LaTeX$, Linux, SQL, MATLAB, HTML5, CSS, JavaScript.

Frameworks PyTorch, TensorFlow, NumPy, DeepSpeed, Git, Anaconda.

Honors and Awards (Selected)

2024 **First Prize**, Sa Shixuan Elite Fund Scholarship

2023 **Finalist**, Mathematical Contest in Modeling (top 2%)

2023 **Best Creativity Award**, Student Entrepreneurship Star Competition, JD Cup (50,000 RMB prize)

2022 **National First Prize**, National College Student Mathematical Modeling Competition (top 0.60%)

2022 **Best System**, GSAI Artificial Intelligence Integrated Design

2022 **1st place**, RUC Programming Course Group OJ Contest (Captain, 1/22)